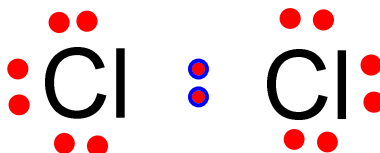


Types of Covalent Bonds

RECALL - Electronegativity is a measure of the ability of an atom to attract electrons in the context of a chemical bond.

Refer to chart on pg 36 of textbook

If two atoms have equal electronegativities (ie the same element), they will form a **non-polar** (or **pure**) covalent bond. They will share the electrons equally.

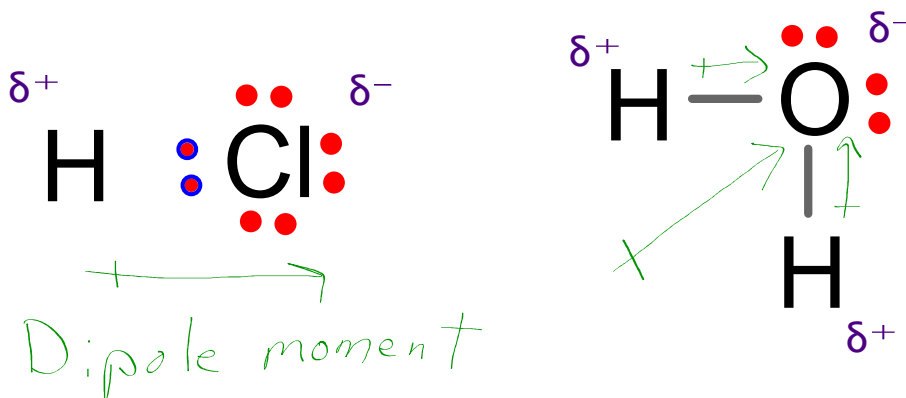


In practice, if the difference in electronegativities are within **0.5**, it is considered a pure covalent bond.

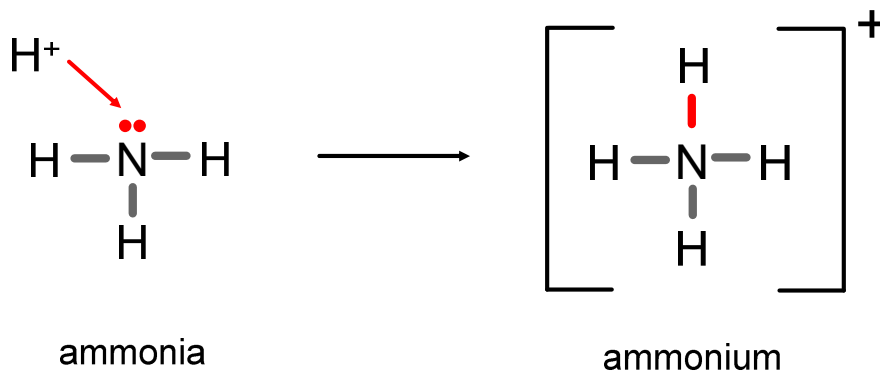


If, however, the two atoms have different electronegativities, they will form a **polar** covalent bond. The electrons are shared *unequally* between the two atoms.

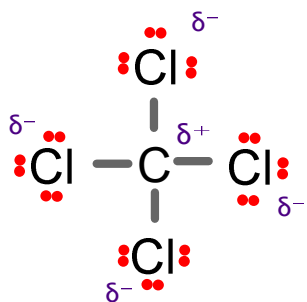
The larger the difference in electronegativities, the more polar the molecule. If the difference is more than **1.7**, then this is indicative of an ionic bond. An electronegativity difference less than 1.7 would be a covalent bond.



The third possibility is **Coordinate Covalent Bonding**. This is a bond in which both of the shared electrons come from the same atom. The most common occurrence of this is seen in ammonium.



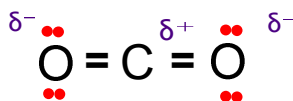
If a molecule is symmetrical, it will be non-polar despite containing polar bonds. Consider carbon tetrachloride, CCl_4



C-Cl bond is **polar**

However, CCl_4 is non-polar.
The polarity is spread out evenly throughout the molecule.

Carbon dioxide is another example of a non-polar molecule:



Read pg 60-62

Do pg 63 #10-12, 14